

REACT NOTES

React JS Notes (Basics to Components)

What is React JS?

React JS is an open-source JavaScript library developed by Meta Platforms for building fast, interactive, and reusable user interfaces. React follows a Component-Based Architecture where the UI is divided into small reusable components. Instead of updating the entire webpage, React updates only the changed part using a Virtual DOM, making applications faster and more efficient. Today, React is widely used for building Single Page Applications (SPA), dashboards, e-commerce applications, CRM systems, admin panels, and enterprise web applications.

Why React?

Before React, developers used pure JavaScript or jQuery to manipulate the DOM directly. As applications grew larger, managing UI updates became difficult and code became harder to maintain. React solves these problems by introducing reusable components, state management, Virtual DOM, and one-way data flow. React helps developers build scalable applications with better performance and maintainability.

Applications Built Using React

Facebook

Instagram

Netflix

WhatsApp Web

Airbnb

Uber Dashboard

Admin Panels

E-Commerce Applications

React Features

Component Based Architecture

Virtual DOM

Reusable Components

One Way Data Binding

JSX

Fast Rendering

Easy State Management

Large Community Support

What is a Component?

A Component is a reusable piece of UI. Instead of writing the same HTML multiple times, React allows developers to create components and reuse them wherever needed. Components make applications modular, maintainable, and easier to scale.

Example

```
function Header(){  
  
  return <h1>Welcome</h1>;  
  
}  
Usage:  
<Header />
```

Types of Components

Functional Component

A Functional Component is a JavaScript function that returns JSX. Modern React applications primarily use Functional Components because they support Hooks and have cleaner syntax.

Example

```
function App(){  
  
  return <h1>Hello React</h1>;  
  
}
```

Class Component

A Class Component is an ES6 class that extends `React.Component`. Before Hooks were introduced, Class Components were used for state and lifecycle methods. Today, Functional Components are preferred.

Example

```
import React,{Component} from "react";
```

```
class App extends Component{

  render(){

    return <h1>Hello React</h1>;

  }

}

export default App;
```

What is JSX?

JSX stands for JavaScript XML. JSX allows developers to write HTML-like syntax inside JavaScript. React converts JSX into JavaScript objects using Babel before rendering it in the browser. JSX makes React code easier to read and write.

Example

```
const element = <h1>Hello React</h1>;
```

Without JSX:

```
const element = React.createElement(
  "h1",
  null,
  "Hello React"
);
```

JSX Rules

Only One Parent Element

Use className Instead of **class**

Use htmlFor Instead of **for**

Every Tag Must Be Closed

JavaScript Uses {}

Example

```
function App(){

  const name = "Sudhan";

  return(

    <div>

      <h1>{name}</h1>
```

</div>

);

}

React Project Setup

Step 1 - Install Node.js

Download and install Node.js.

Check Version:

node -v

npm -v

Step 2 - Create React Project

Using Vite:

npm **create** vite@latest

Project Name:

react-app

Select:

React

JavaScript

Step 3 - Move Into Project

cd react-app

Step 4 - Install Packages

npm install

Step 5 - Run Project

npm run dev

Output:

http://localhost:5173

React Project Structure

react-app

|

├─ node_modules

├─ public

└─ **src**

```
|  ├── assets
|  ├── App.jsx
|  ├── main.jsx
|
|── package.json
|── vite.config.js
```

main.jsx

main.jsx is the entry point of a React application. It loads the App component and renders it inside the root element.

Example

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
```

```
ReactDOM.createRoot(
  document.getElementById("root")
).render(
```

```
  <App />
```

```
);
```

App.jsx

App.jsx is the root component of the application. All other components are generally rendered inside App.jsx.

Example

```
function App(){
```

```
  return(
```

```
    <h1>
```

```
      Welcome To React
```

```
    </h1>
```

```
  );
```

```
}
```

```
export default App;
```

Import & Export

React uses modules to share components between files.

Export

```
export default Header;
```

Import

```
import Header from './Header';
```

Why React Uses Components?

Components allow:

Code Reusability

Easy Maintenance

Scalability

Modular Architecture

Better Project Structure

Instead of writing the same navbar, footer, or card multiple times, React components can be reused throughout the application.

React Rendering Process

User Opens Application

↓

main.jsx

↓

App Component

↓

Child Components

↓

JSX Converted **To** JavaScript

↓

Virtual DOM Created

↓

Real DOM Updated

↓

UI Displayed

Interview Questions

What is React?

React is a JavaScript library used for building reusable and interactive user interfaces.

What is a Component?

A Component is a reusable piece of UI that returns JSX.

What is JSX?

JSX is a syntax extension that allows HTML-like code inside JavaScript.

Difference Between Functional and Class Components?

Functional Component → JavaScript **Function**

Class Component → ES6 **Class**

Why React is Fast?

Because React uses Virtual DOM and updates only the changed elements instead of reloading the entire page.

What is Virtual DOM?

Virtual DOM is a lightweight copy of the Real DOM used by React to efficiently update the UI.

What is the Entry Point of React?

main.jsx

What is the Root Component?

App.jsx